



Design and evaluation of a privacy-preserving multi-level federated learning architecture for airport biometric check-in

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ABSTRACT

The rapid adoption of automated airport check-in systems using facial recognition raises significant privacy concerns due to their reliance on centralized deep learning models that store and transmit biometric data from edge devices. While Federated Learning (FL) is a promising approach for privacy preservation, its effectiveness in biometric identification remains underexplored, particularly in real-world environments like airports. This study assesses the privacy implications of FL in facial recognition by comparing three architectures. A first centralized system, where biometric data is sent to a central server for model training and inference, posing significant privacy risks. The second is a one-level FL architecture, where biometric data remains on local devices, and only model updates are shared with a central aggregator. The third is a two-level FL architecture, introducing an additional aggregation layer among airlines to enhance model generalization while preserving privacy. To ensure a rigorous privacy preservation evaluation, we integrate both quantitative and qualitative metrics. For the quantitative assessment, we leverage the Privacy Meter Tool, which enables simulations of Membership Inference Attacks and the application of Differential Privacy as a mitigation technique. For the qualitative evaluation, we conduct a Data Protection Impact Assessment to analyze potential privacy risks from a regulatory perspective. Additionally, we assess model accuracy, computational efficiency, and communication overhead to determine FL's feasibility in large-scale airport environments. Our results show that while FL significantly reduces privacy risks, the two-level FL approach introduces new vulnerabilities, such as model poisoning risks and privacy-utility trade-offs, requiring further mitigation strategies like DP.

1. Introduction

In recent years, the adoption of biometric data to identify people has become increasingly prevalent in several sectors, including security, health care, and transportation, with the installation of technology systems in different settings [1].

In fact, stringent demands for greater safety and security, such as in the cases of border control, national ID cards, and government access systems, have made traditional identification methods no longer adequate. The latter use credentials such as documents, passwords, and PINs which, however, can be lost or stolen. For these reasons, biometric recognition systems have been increasingly adopted [2]. This kind of system exploits biometric data, which are based on a person's unique and persistent physiological features.

Technologies such as facial recognition, retinal scanning, fingerprint analysis, and even the fusion of such data are now integrated into mul-

tipale systems, applications, and devices to increase the compactness of information and consequently improve the efficiency of automatic identification processes [3,4].

Facial recognition is the most commonly utilized technique since the invention of devices with identification functions. Indeed, smartphones and laptops are now equipped with sensors capable of collecting facial features, which replace the use of a PIN as a screen lock. The latter is part of a very large family of applications of facial recognition in the field of security and control. More sophisticated are device installations for controls in video surveillance scenarios [5] and access in private areas [6], such as buildings or computer systems. Thus, face recognition has become a normal and daily activity also in many airports around the world [7]. Specifically, the air transportation industry has seen an increasing implementation of biometric systems to optimize passenger flow at airports, and reduce waiting times while increasing security in and around airports [8]. New boarding systems with biometric controls

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can be found, for example, at San Francisco International Airport, where *Smart Path technology*¹ [9] has been in operation since 2021, or even the *FaceBoarding service*² at Milan Linate airport, where travelers are free to choose whether or not to use the new system; and a further example involves the *Smart Travel project* being carried out at the airports in Abu Dhabi.

Biometric systems can be categorized as complex systems in that they rely on the use of devices sensitive to a certain type of biometric data (e.g., retinal detection requires a camera but voice detection operation requires a microphone), for the implemented algorithms, which are divided into verification methods and identification methods [10], and by their installation and structures (they are used to create intelligent and connected environments, such as smart cities). Being complex systems, these must adapt to technological innovations, and so the evolution of biometric systems is strictly linked to the use of **Artificial Intelligence (AI)** [11]. As explained in [12], **Deep Learning (DL)** methods have demonstrated considerable success for learning tasks in domains, like computer vision and **Natural Language Processing (NLP)**, which are closely related to face and speaker recognition tasks, increasing the impact of **AI** in the field of biometrics.

While the integration of **AI** into biometric systems, such as **DL** algorithms to perform facial recognition of automated check-ins improves the efficiency and accuracy of identification processes [13], it raises important concerns related to the privacy of personal data [14] and the use of **AI** algorithms may pose risks of unauthorized surveillance and data misuse. Problems that have been extensively analyzed in the implementation of *Smart Path* and *FaceBoarding* systems. In fact, airport security relies on biometric surveillance, requiring the capture and centralization of biometric data in extensive databases, which increases exposure to potential cyber-attacks and unauthorized access.

Traditional **AI** models for biometric recognition often rely on centralized architectures, where biometric data collected from multiple devices are transmitted to a central server for processing and model training. This centralization introduces several vulnerabilities. In general, using centralized servers carries many risks, including a) scalability and latency, b) cyberattacks and unauthorized access, and c) Single Point of Failure. In the first case, biometric systems must handle millions of simultaneous requests, which can cause bottlenecks in recognition operations and is a critical problem in airports, where rapid identification is needed. Moreover, in the second case, a single server is a target for hackers because it contains a large amount of sensitive data and has a unique vulnerability point.

Security protocols frequently leverage the combined power of **AI** and biometrics. **AI** algorithms analyze data to reveal underlying patterns, while biometrics capture the unique traits of each individual. Together, these technologies create a profile of user behavior that enables differentiation between individuals. In [15] a survey is proposed concerning the potential attacks to which a user authentication system based on these tools is exposed, and the principal limitations that are still unsolved are described. These risks highlight the need for alternative approaches that reduce the dependence on centralized data storage while maintaining the efficiency and accuracy of biometric recognition systems.

Addressing these concerns, recent studies explore privacy-preserving techniques, including **Federating Learning (FL)**, advanced data **Homomorphic Encryption (HE)**, and database security [16].

This work addresses the issue of privacy preservation in automated airport check-in systems based on facial recognition. It explores the main vulnerabilities, mitigation strategies, and the role of **FL** in safeguarding biometric data. Two decentralized **FL**-architectures are proposed: the first involves a network of edge devices used to identify passengers of a single airline through a central aggregator; the second extends this model to enable collaboration among multiple airlines, where

each airline operates as an edge aggregator within a two-level **FL** hierarchy. For comparison, a traditional centralized architecture is also considered. The research aims to evaluate how different architectural configurations—centralized, one-level **FL**, and two-level **FL**—affect privacy leakage, model accuracy, and system scalability. The main contributions include: (i) The implementation of a comparative framework to analyze privacy risks in three biometric identification architectures using the Privacy Meter tool; (ii) The introduction of a novel two-level **FL** architecture that enables multi-airline collaboration while enhancing privacy preservation; (iii) The application of both quantitative (via **Membership Inference Attack (MIA)**) and qualitative (via **Data Protection Impact Assessment (DPIA)**) methods to evaluate the trade-offs between privacy and model performance.

The paper is organized as follows. After the introduction, **Section 2** reviews privacy and **FL** concepts in literature, and then **Section 3** details the description of the proposed architecture. **Sections 4** and **5** present the experimental setup and a discussion of the relative results. Finally, **Section 6** makes the final observation.

2. Related works

Face recognition has emerged as a pivotal technique in biometric systems, enabling the identification or verification of individuals through facial features extracted from images or videos [7]. Moreover, advances in sensor and imaging technologies have expanded their applications in **Machine Learning (ML)** and **DL** [17]. The works in [17,18] provide a comprehensive overview of face recognition methodologies, analyzing a broad spectrum of classical and **DL**-based techniques applied across multiple datasets. These studies expand the application of facial recognition and emphasize performance-related challenges such as pose and illumination variation, occlusion, and aging effects. On the other side, **Convolutional Neural Networks (CNNs)** [19] effectively addresses those challenges with recent advancements like attention mechanisms and adaptive loss functions, enhancing their robustness and accuracy. In contrast, the Vision Transformer (ViT) [20] model introduces a novel approach by adapting the Transformer architecture from **NLP** to image classification. Moreover, DeepFace [21] achieved near-human performance on **Labeled Face in the Wild (LFW)** using a 3D alignment step and a nine-layer **CNN**. FaceNet [22] employed triplet loss for accurate face verification via Euclidean embeddings. Real-time face recognition must address not only computational constraints but also privacy concerns, especially with regulations like **General Data Protection Regulation (GDPR)**, which enforce data anonymization, minimization, and user consent. Privacy-preserving techniques include **Differential Privacy (DP)** [23], **FL** [24,25], or applications for biometric systems [26]. As in [27], which presents a biometric authentication for banking systems based on **HE**, or the recent contribution in the healthcare domain [28], which designed a **FL** framework for multi-cloud environments that combines **HE**, adaptive **DP**, and secure multi-party computation to ensure privacy preservation across heterogeneous datasets. While demonstrate strong privacy guarantees, the works do not explore decentralized computation or privacy auditing, limiting their applicability in privacy-sensitive environments. **Table 1** summarizes strategies discussed above.

In contrast, our work focuses on privacy-preserving facial recognition in airport check-in scenarios. To the best of our knowledge, no prior study provides a quantitative comparison across **FL** architectures supported by privacy auditing tools as Privacy Meter.

3. Proposed architecture

Automated airport check-in systems based on facial recognition are becoming widespread, posing privacy concerns like unauthorized access, data leaks, and model inversion attacks.

FL provides a compelling alternative to centralized approaches by enabling collaborative model training without transferring raw biometric data [29]. This paradigm prevents direct sharing of facial images

¹ <https://www.infrajournal.com/>

² <https://www.milaninate-airport.com/en/voli/face-boarding>

Table 1
Summary of works on privacy-preserving face recognition and federated learning.

Reference	Domain / Scenario	FL / DP / HE	Privacy Auditing	Architecture Type	Notes
[18]	Face recognition	x	x	–	Traditional and DL methods; issues like occlusion and pose variation.
[17]	Face recognition	x	x	–	DL techniques focused on performance and technical evolution.
[27]	Banking biometrics	HE	x	Centralized	Biometric authentication using HE; lacks decentralization and privacy auditing.
[26]	Biometric systems (general)	FL, DP, HE	x	–	Privacy-preserving methods; no architectural or quantitative evaluation.
[28]	Healthcare / multi-cloud	DP, HE	x	Federated, multi-cloud	Architecture for cloud integration, no privacy auditing or biometric-specific analysis.
Our Work	Airport check-in biometrics	FL, DP	Privacy Meter	Centralized, One-Level and Two-Level FL	Quantitative comparison across FL architectures in biometric context; privacy audits and risk mapping.

while ensuring that different entities - such as airlines and airport authorities - can collaborate on a shared biometric authentication model without violating privacy regulations such as [GDPR](#). However, existing [FL](#) deployments often rely on a flat architecture where edge devices directly communicate with a central server. Our approach introduces a hierarchical architecture that explicitly reflects the multi-stakeholder structure of real airports, enabling per-airline model aggregation and cross-airline coordination. [FL](#) mitigates the risk of data breaches and unauthorized access but introduces challenges related to privacy, security, and accuracy.

To explore the trade-off between accuracy and privacy, the case study compares three architectures centered on key components such as passengers and edge devices, each with distinct operational and privacy characteristics. At the edge devices level, passengers represent the primary entry point of biometric data, where facial images are captured and preprocessed. These edge devices, such as airport kiosks, ensure that data processing occurs, reducing exposure to external threats. The Airline Server (Local Aggregator) functions as an intermediary in the [FL](#) framework, where the aggregation of models occurs. In the Two-Level [FL](#) approach, an additional aggregation layer, the Global Aggregator (Airport-Level Aggregator), integrates model updates from multiple airline servers. This layer is intended to enhance the model's overall robustness by mitigating the bias that arises from training on data from a single airline while preventing the direct exchange of data.

Beyond the privacy concerns, each architectural approach impacts system performance, scalability, and operational feasibility. Centralized models generally yield higher accuracy due to access to a more extensive dataset; however, they also introduce a *single point of failure*. [FL](#) models distribute computation across multiple nodes, reducing privacy risks but suffering from heterogeneous data distribution and increased communication overhead. Furthermore, adversarial threats, such as [MIA](#) and membership inference attacks, vary across these architectures.

In light of potential issues and suggested solutions, this section aims to give a thorough understanding of the benefits and drawbacks of each strategy in the context of privacy-compliant facial recognition for airport check-in systems.

3.1. Privacy threats in biometric systems

The case study presents significant privacy challenges because of the centralizing risks associated with biometric data. Traditional biometric systems store data centrally, making them vulnerable to breaches and misuse.

[FL](#) has been proposed as a privacy-preserving alternative that allows [DL](#) models to be trained on edge devices without directly transmitting biometric data to a central server. However, [FL](#) is not immune to privacy

threats. In literature, different studies have shown that even without access to raw data, attackers can use gradient updates or model parameters to infer information about training data. This means that an adversary could reconstruct sensitive biometric data, determine whether a particular passenger's face was used for training, as discussed in [30,31]. It is important to note that the analysis carried out in this paper is focused on privacy preservation issues rather than cybersecurity aspects.

An example of a privacy risk is [MIA](#) attack, when an adversary can infer training data from a target [ML](#) model since the data contain sensitive information [32]. [MIA](#) raises serious concerns under the [GDPR](#), which defines biometric data as a special category of data and imposes strict requirements on its processing. While [FL](#) reduces direct data sharing, [MIA](#) can still indirectly expose personal information, raising issues [33,34], making [FL](#) architectures vulnerable to [MIA](#). In the case of a one-level network, a malicious agent could use a [MIA](#) to compromise an edge device to gain access to model updates. In this way, the agent could track how model updates change and infer whether specific passengers' data is included in the training data [35].

From a mathematical perspective, given a trained model f_θ and an input sample x , an adversary aims to determine whether x was part of the training set $\mathcal{D}_{\text{train}}$. This can be formalized as a binary hypothesis test:

$$H_0 : x \in \mathcal{D}_{\text{train}} \quad \text{vs.} \quad H_1 : x \notin \mathcal{D}_{\text{train}}.$$

The attacker computes a membership score $s(x)$ (e.g., loss, confidence, gradient norm) and applies a likelihood ratio test or threshold-based classifier to infer membership. The success is evaluated via the ROC curve and its Area Under the Curve (AUC-ROC), where $\text{AUC} > 0.5$ indicates privacy leakage.

On the other hand, in the two-level network, the [MIA](#) malicious agent could be one of the airlines involved in the [FL](#) network, thus becoming a case for model poisoning. In particular, a malicious airline would perform a [MIA](#) by carefully observing how the local model changes over time. Specifically, the airline trains its local model by modifying its dataset by excluding a particular passenger's biometric data, saving the model's weights before sending updates. After that, it re-training the local model by adding the target passenger's biometric data to its local training set. Finally, the new model updates are compared with the previous model updates to check for differences. The airline submits both model updates to the [FL](#) global coordinator. Therefore, if the updates change significantly due to the addition of passenger data, this suggests that the global model is sensitive to that particular passenger's information. In addition, the airline can infer whether that passenger's data has been used by another airline. This risk is amplified due to gradient-based leakage, as shown by the work on white-box [MIA](#) [36].

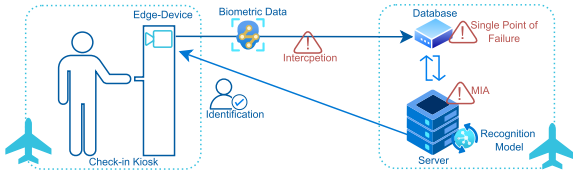


Fig. 1. Centralized network.

Another example of privacy risks is the *Model Inversion Attacks*, which attempt to reconstruct biometric data from the model parameters. In a FL environment, attackers may try to exploit shared gradients to extract sensitive features, even if they do not have access to the model. Recent studies show that FL models trained on facial recognition data are vulnerable to gradient inversion techniques, which allow an attacker to reconstruct images from the weight updates shared during FL training. An attacker could reverse engineer the training process to generate a visual representation of a passenger, as shown in [37]. Additionally, by using *Generative Adversarial Networks (GANs)* techniques, an attacker can improve the quality and fidelity of the reconstructed images [38]. In airport check-in systems, in a FL or non-FL architecture, this attack could allow a malicious entity to recover partial or full reconstructions of facial images. Consequently, raising concerns about identity theft, unauthorized tracking, and biometric fraud.

The *Model Poisoning Attack* functions differently from the prior two attacks. Indeed, it injects malicious updates into the global model in two distinct ways: (1) manipulating local training data to introduce biases or degrade model performance; and (2) manipulating the labels of faces in the local dataset to confuse the global model and result in the misclassification of users. For instance, in the case of airport security, an attacker could train a model with a trigger condition (e.g., wearing a specific hat or sunglasses) that causes the system to incorrectly identify the person as an authorized passenger.

3.2. Proposed architectures

3.2.1. Centralized network

In this case, facial recognition is performed using a recognition model hosted on a centralized server. The entire process, including data collection, model training, and identity verification, is based on this server. As shown in Fig. 1, devices collect and transmit the data to the central dataset. The central server uses the data for two core operations: identity verification, to perform automated check-in and provide the outcome of passenger identification in real-time, and training recognition model, to train or update the DL model. The centralized nature of this approach allows for efficient model updating and high computing power, but it may raise privacy issues.

A prior problem is the creation of a *single point of failure*, making the system highly vulnerable to data breaches and unauthorized access. Another risk could be *MIA*. In this scenario, an adversary exploits answers to model queries to determine whether a given passenger's data was used during training.

3.2.2. One-level federating learning network

The second scenario is shown in Fig. 2. FL is introduced to mitigate the centralized network's risks. Each airline operates its independent FL system, which does not communicate with those of other airlines. The network performs the two main operations, real-time identification and model training, on edge devices, which do not need to transmit biometric data to the central dataset since the facial recognition model is trained locally. Only local model weight updates are shared with the FL aggregator to refine the global model before redistributing it to edge devices for further local training. This means that communication between the edge devices and the aggregator involves model weight updates, and no biometric data is transmitted.

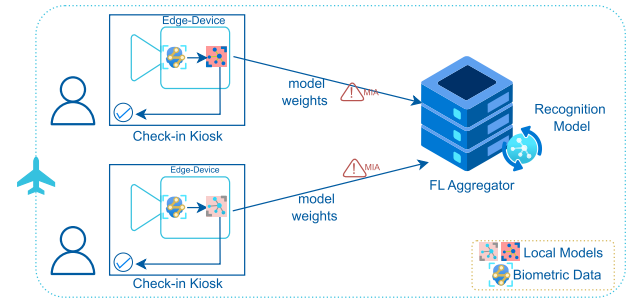


Fig. 2. One-level network.

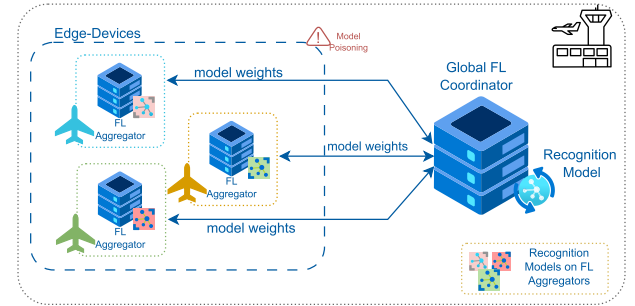


Fig. 3. Two-level network.

A key advantage is the architecture's ability to enhance passenger privacy by reducing the risk of unauthorized access. Additionally, the system remains operational even in case of intermittent connectivity, improving robustness. The risk lies in the exchange of weight updates from the local training; in fact, adversaries could attempt *MIA* by analyzing updates sent from edge devices to the FL aggregator.

3.2.3. Two-level federating learning network

This scenario represents a two-level FL network, designed to enhance privacy and model generalization by enabling collaboration between multiple airlines without directly exchanging sensitive biometric data. The system, built on the one-level FL architecture, introduces an additional layer of collaboration at the airline level. As illustrated in Fig. 3, the architecture consists of two components: (1) a global FL coordinator and (2) a set of edge devices corresponding to different one-level FL systems.

At the one-level networks, each airline independently manages its identity operations through edge devices, which locally train the facial recognition model. These edge devices periodically transmit updated model weights to their respective airline aggregator. At the two-level network, these airline aggregators act as edge devices within the global FL system. Each aggregator communicates with the global coordinator, exchanging aggregated model parameters instead of biometric data. The coordinator applies secure aggregation techniques to refine the global model, ensuring privacy-preserving updates.

The main advantage is that each airline maintains full control over its passengers' information while still benefiting from a broader knowledge base. However, a major risk could be model poisoning, when a malicious airline or compromised aggregator injects manipulated updates to introduce biases into the global model.

3.3. CNN setting for facial recognition task

To perform facial recognition, the three architectures leverage a CNN, in Fig. 4. Specifically, the CNN is built using PyTorch and consists of three layers designed for feature extraction and classification, trained on the LFW dataset.

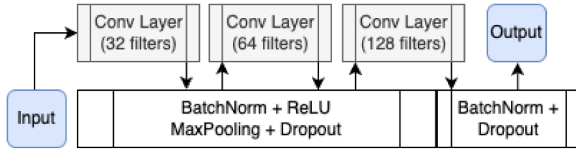


Fig. 4. CNN architecture.

The first one applies a 3x3 convolution with 32 filters, the second layer is modelled with 64 filters, finally the third layer, with 128 filters, refines the feature representation. Each layer is followed by batch normalization, ReLU, max pooling (22), and dropout (25 %). The model uses Cross-Entropy Loss, Adam optimizer (learning rate 0.0005, weight decay 0.0001), and was trained for 100 epochs with batch size 64.

3.4. Mitigation strategies

This section explores two key strategies to improve privacy preservation in FL biometric systems during model training and aggregation.

The first is **Secure Aggregation (SA)** a cryptographic protocol designed to protect individual model updates in FL by ensuring that only aggregated model updates are accessible to the central aggregator [39,40]. This technique aims to prevent adversaries from reconstructing individual data, mitigating the risk of MIA and model inversion attacks. Let $\Delta\theta_i$ denote the model update computed by client i during a given round. In SA, each $\Delta\theta_i$ is encrypted or masked before being sent to the server. The aggregator can only reconstruct the aggregated value:

$$\sum_{i=1}^N \Delta\theta_i,$$

but cannot access any individual $\Delta\theta_i$. This is typically achieved through additive masking protocols or cryptographic techniques such as HE or secret sharing. As a result, even a compromised aggregator cannot infer any participant's specific update. In the context of one-level FL architecture, SA could be implemented at the airline level, preventing the airline's aggregator from accessing the raw updates of individual edge devices. In the two-level FL architecture, SA could be extended to the global aggregation layer, ensuring that no airline can extract information about others' model updates. However, SA introduces computational and communication overhead [41].

Another technique for preserving privacy is **DP**, El Ouadrhiri and Abdelhadi [42], which provides a solid mitigation against MIA by injecting calibrated noise into model updates, thus preventing attackers from distinguishing whether a particular individual's data was used during training.

Formally, a randomized algorithm $\mathcal{A}$ satisfies (ϵ, δ) -DP if, for all neighboring datasets D, D' differing in a single sample, and for all measurable subsets S of outputs:

$$\Pr[\mathcal{A}(D) \in S] \leq e^\epsilon \cdot \Pr[\mathcal{A}(D') \in S] + \delta,$$

In our setting, we apply the Gaussian Mechanism, where gradients are first clipped to have bounded ℓ_2 norm C , and then perturbed with noise drawn from the distribution $\mathcal{N}(0, \sigma^2 C^2 I)$, where σ controls the noise level.

Privacy accounting mechanisms are used to track the cumulative privacy loss ϵ over the course of model training.

In the one-level architecture, DP can be implemented at the edge device level, before the transmission of updates to the airline's aggregator. On the other hand, in the two-level architecture, DP mechanisms can be integrated at the FL coordinator level, applying noise to the aggregated updates. However, the implementation of DP introduces a trade-off between privacy and usefulness, as too much noise may compromise the accuracy of the model [43].

4. Experimental setup

4.1. Dataset, tools, and integration

Labeled Face in the Wild Dataset. Introduced by the author in [44], the LFW is a collection of face photographs designed to study the unconstrained recognition problem. The uniqueness of this database is that provides a set of photographs of faces under conditions encountered in everyday life. Thus, the database has “natural” variability in factors such as pose, lighting, accessories, occlusions, and background.

The dataset comprises 13233 images of 5749 individuals, with 1680 subjects having multiple images. Moreover, each image is labeled with the name of the person depicted, thus enabling both *face verification* (determining whether two images represent the same individual) and *face recognition* (identifying the person in an image from a known set of individuals). The images are all 250x250 pixels, but the size in KB may differ, captured in RGB, and saved in jpg file format.

For the experimental setup, LFW is integrated into our pipeline through the Scikit-learn³ site, as during the setting of the experiments, the original site⁴ was found to be unsearchable.

By leveraging the LFW dataset, our study benefits from a rich set of facial images, enabling us to develop and evaluate facial recognition models under conditions that mirror real-world scenarios.

Tools and Frameworks. To implement and evaluate the architectures of Section 3.2, different tools and libraries are exploited. The main focus is to analyze privacy in different scenarios, in particular, to compare privacy loss between one- and two-level FL architectures, and the tool used for the objective is the Privacy Meter Tool.⁵

Privacy Meter is an open-source library designed to analyze the privacy risks associated with ML models. It provides methods to assess MIA, gradient leakage, and other privacy vulnerabilities. It was chosen because it is a valid tool for testing data privacy in a wide range of ML algorithms and aims to assess MIA risks in FL architectures by analyzing whether an attacker can infer training membership from model updates. In addition, the open-source DL library PyTorch and the open-source framework designed for implementing and managing FL workflows OpenFL, were crucial tools in structuring the three architectures. In this study, PyTorch is used for implementing the CNN, which is used as the core of all three structures. On the other hand, OpenFL is utilized to organize, simulate, and evaluate both the one-level and two-level architectures.

Customization and Integration. Privacy Meter Tool, by default, supports several datasets and models but allows customization of both elements. Therefore, we integrated the dataset LFW within the Tool by pre-processing the data to align it with Privacy Meter requirements. In fact, the images were converted to tensor-based representations but were not resized, as this operation would have affected recognition performance. However, it is important to emphasize that facial recognition performance is not our goal. Privacy Meter was then employed to analyze the membership inference vulnerability of the trained models by simulating adversarial attacks on the dataset.

4.2. Quantitative evaluation of privacy risks

4.2.1. Centralized architecture

In this experiment, we evaluate the face recognition performance of the CNN trained in the centralized setting of Section 3.2.1, on the LFW dataset. The results are visualized in Fig. 5(b) (Training and Test Loss) and Fig. 5(a) (Training and Test Accuracy).

³ https://scikit-learn.org/0.19/datasets/labeled_faces.html

⁴ <http://vis-www.cs.umass.edu/lfw/>

⁵ https://github.com/privacytrustlab/ml_privacy_meter

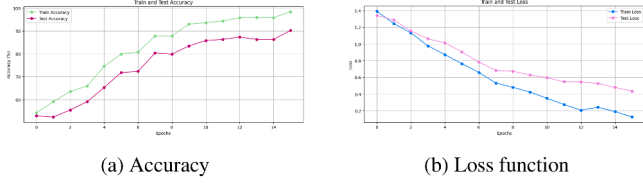


Fig. 5. Training and test accuracy and loss.

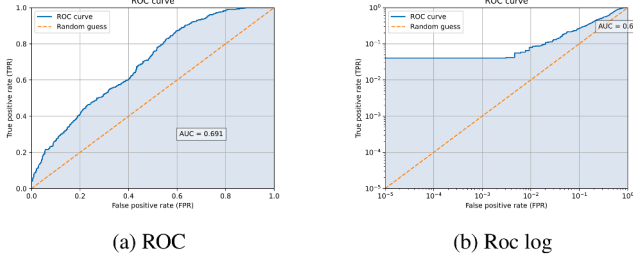


Fig. 6. ROC curve for MIA experiment.

Fig. 5(b) indicates that the model effectively learns the facial features over time, the Test Loss follows a decreasing trend showing that the model learns the task. Similarly, the training and test accuracy curves (Fig. 5(a)) confirm the model's strong generalization ability.

Privacy Audit. Regarding the privacy audit experiment associated with our trained models, we used our modified version of the Privacy Meter Tool. In particular, the privacy audit experiment evaluates the vulnerability of ML models to MIA.

The audit pipeline [32] is structured as follows: 1) *Dataset preparation*, The dataset is split into training and testing; 2) *Model Training and Selection*, the CNN is trained and the model is evaluated for privacy leakage; 3) *MIA Setup*, we use *Robust Membership Inference Attack (RMIA)* [45]; and, 4) *Attack and Evaluation*, the attacker queries the trained model and assigns a membership score. The *Receiver Operating Characteristic (ROC)* curve and the *Area Under the Curve (AUC-ROC)* metrics are used to evaluate the effectiveness of the attack. AUC-ROC values above 0.5 indicate privacy leakage, with higher values suggesting stronger model vulnerability.

In formal terms, the attack on the membership inference can be modeled as a hypothesis test with two hypothesis: a) $H_0 : x$ is part of the training set (IN world), and b) $H_1 : x$ is not part of the training set (OUT world). Thus, an adversary attempts to distinguish between these two worlds based on the model's output distribution. The RMIA uses an improved method of the traditional likelihood ratio test, calibrating membership probability with reference models:

$$LR(f, x) = \frac{P(x|f)P(x)}{P(z|f)P(z)}$$

where x represents a potential member and z a non-member extracted from the population [46].

The ROC curves, in Fig. 6(a) and (b), illustrate the effectiveness of MIA against the trained model, as evaluated by the Privacy Meter Tool. The AUC-ROC value of 0.691 indicates that the attacker performs better than random guessing (The AUC-ROC = 0.5), meaning the model leaks some membership information. A higher The AUC-ROC suggests stronger vulnerability to MIA.

In conclusion, the experiment with the centralized CNN establishes a benchmark for the FL-based architectures, where privacy constraints may impact accuracy. Indeed, these results will be compared to FL experiments to assess whether privacy-preserving methods affect recognition accuracy.

Table 2

Performance summary of one-level federated learning experiment.

Clients	Avg. Aggregated	Avg. Training Loss	Avg. Local Validation
2	0.8039	0.4141	0.7884
5	0.6583	0.7665	0.6291

4.2.2. One-level federated learning experiment

In the experiment for the one-level FL scenario, the edge devices act as FL nodes, enabling local training and subsequent aggregation in the central server, as described in Section 3.2.2.

This experiment used the Privacy Meter Tool to assess privacy risks from MIA to model updates in FL. The experiment used OpenFL, modified to integrate the Privacy Meter Tool. The use of LFW dataset made the simulation particularly challenging, as for some classes (i.e., some public people) there were few images. For this reason, it was decided to use only those classes for which the dataset stores more than 70 images reducing the total number of images, upsetting other challenges to face.

In this experiment the same CNN was used to assure comparability across experiments. This took precedence over seeking pure performance in terms of accuracy, which is not the purpose of our proposal.

The OpenFL configuration and integration were completed with the following steps: 1) *Local Training*, in each round of FL, each edge device trains its local model using the dataset available to it, and the model is initialized with the current global model; 2) *Privacy Audit*, after training, both the current global model and the updated local model are analyzed by a privacy audit module and the module generates a privacy loss report; finally, 3) *Model aggregation*, local updates are sent to the aggregator.

Privacy was assessed using the *Population Attack* from Privacy Meter Tool and integrated into OpenFL. An adversary computes signals (e.g., loss or logits) on a disjoint dataset to define a threshold minimizing false positives, then uses it to distinguish training from test samples.

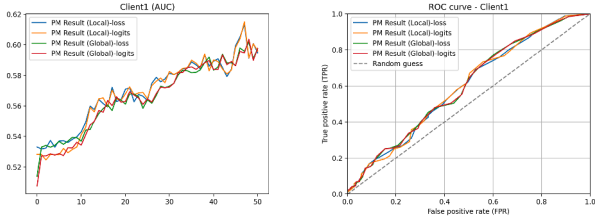
To integrate this attack into the privacy audit, it is assumed that there is a separate dataset dedicated to the audit, the size of which is defined by a specific parameter in the code. The system also allows the user to choose which signal to use to determine the possible membership of a given sample in the training set. Options include loss, logits, and gradient norm.

The success of the attack is evaluated via the ROC curve and monitoring the AUC-ROC during training. In addition, the system allows the user to set a tolerance threshold for the false-positive rate to obtain a privacy report showing the maximum true-positive rate, i.e., the percentage of training data that were correctly identified.

Experimental results and Privacy Audit. The experiments were conducted in two different ways to compare results. The first with two clients as edge nodes of FL architecture and the second one with five nodes. In both cases, we set 50 rounds of FL training and 10 local training epoch ran on each edge device before every aggregation round. The optimizer used in the CNN was the *Stochastic Gradient Descent (SGD)* with a learning rate set to 0.0005, to balance convergence speed and model stability. Table 2 summarizes the results obtained for both scenarios.

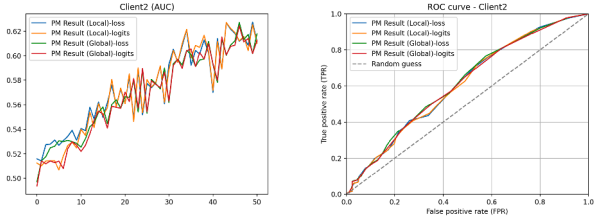
The privacy audit results highlight the mitigation effect of the FL architectures on MIA, in both scenarios. The analysis of the AUC-ROC and ROC curves, specifically for each scenario, allows us to obtain more specific insights.

Figs. 7 and 8 show the trends of AUC-ROC and the ROC curve at the end of all FL training rounds. The AUC-ROC curves show a gradual increase over the training rounds, indicating that the attacker obtains more discriminative power as the model is trained and the ROC curves support this observation. This suggests that even in a FL environment, there is still information leakage from model updates, making membership inference possible, although considerably mitigated compared to the centralized architecture.



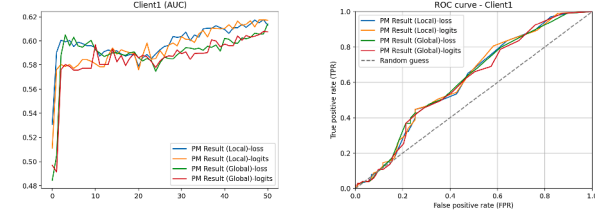
(a) AUC for Client 1

(b) ROC at 50 Rounds for Client 1

Fig. 7. Privacy auditing - Client 1 in 2 nodes FL scenario.

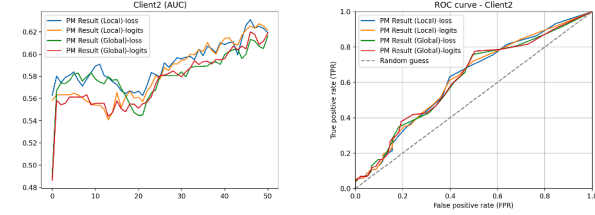
(a) AUC for Client 2

(b) ROC at 50 Rounds for Client 2

Fig. 8. Privacy auditing - Client 2 in 2 nodes FL scenario.

(a) AUC - Client 1

(b) ROC - Client 1

Fig. 9. Privacy auditing - Client 1 in 5 nodes FL scenario.

(a) AUC - Client 2

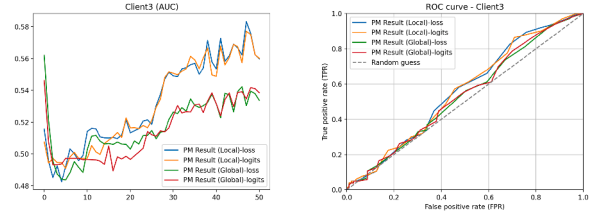
(b) ROC - Client 2

Fig. 10. Privacy auditing - Client 2 in 5 nodes FL scenario.

Increasing the number of nodes to five in the FL architecture a significant reduction in privacy leakage is obtained. The AUC-ROC values remain lower than the 2 client scenario and the ROC curves reflect this improvement, from Figs. 9–13. Thus, increasing the number of FL participants introduces variability in model updates, reducing the MIA. The attacker's performance approaching the random guess line more closely, as shown by ROC curves of all nodes, indicating an increment of difficulty for the attacker to distinguish between training and non-training data, reinforcing the idea that larger FL networks provide a defense against MIA.

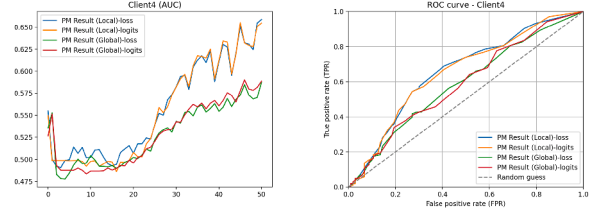
4.2.3. Two-level federated learning experiment

In this experiment, DP is applied to local model updates as an additional privacy-preserving layer before the second-level FL aggregation. DP injects calibrated Gaussian noise into gradients to obfuscate individual updates. To preserve the simulation validity, we focused on the local



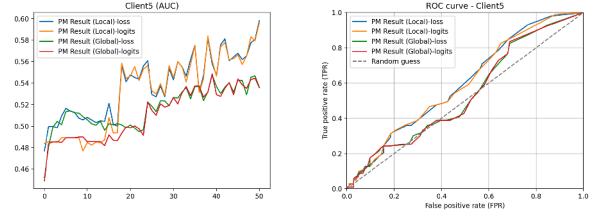
(a) AUC - Client 3

(b) ROC - Client 3

Fig. 11. Privacy auditing - Client 3 in 5 nodes FL scenario.

(a) AUC - Client 4

(b) ROC - Client 4

Fig. 12. Privacy auditing - Client 4 in 5 nodes FL scenario.

(a) AUC - Client 5

(b) ROC - Client 5

Fig. 13. Privacy auditing - Client 5 in 5 nodes FL scenario.

DP effects rather than reimplementing the entire FL pipeline. Opacus,⁶ a PyTorch-based library for DP, was integrated into the training workflow, together with an adapted version of the Privacy Meter tool used to evaluate the effective privacy budget under membership inference auditing.

The pipeline used to run this experiment, inspired by [47], includes: 1) the baseline model training without DP; 2) training with DP using the Gaussian Mechanism with noise addition to model gradients before aggregation; 3) Privacy Auditing using Privacy Meter under MIA; and, 3) Performance Evaluation comparison between the non-DP and DP models.

A DP experiment takes as input a randomized algorithm and a predefined confidence tolerance, producing a statistically robust lower bound $\hat{\epsilon}$ for the true DP parameter. This bound is established with high confidence, ensuring that the probability of $\hat{\epsilon}$ underestimating ϵ remains at most β , given that the algorithm T adheres to (ϵ, δ) -DP. The design of such auditing procedures typically leverages the fundamental constraint of (ϵ, δ) -DP on the statistical power of binary membership hypothesis tests, as in [48,49].

Experimental Results and Privacy Audit. The two experiments, the first one without DP and the second one training the model with Gaussian DP, were repeated 4 times to increase reliability and reduce unpredictability. The results, in terms of performance, were very good for the first experiment establishing the average value of training accuracy around 0.97 and accuracy around 0.84. The results of the second one

⁶ <https://opacus.ai/>

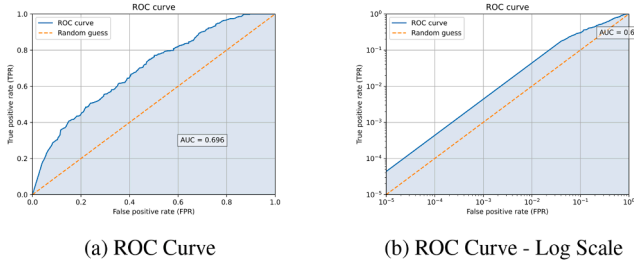


Fig. 14. Non-DP scenarios.

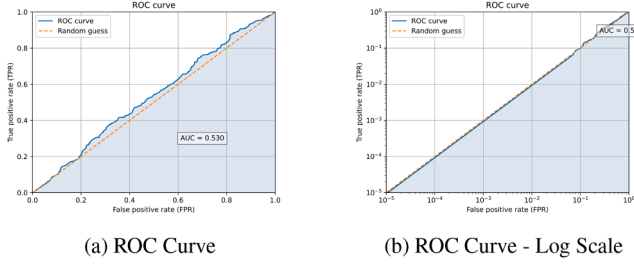


Fig. 15. DP scenarios.

demonstrated an important decline establishing an accuracy value of 0.64 and an $\epsilon = 121$.

The privacy risk evaluation is shown by the evaluation of MIA with ROC and AUC-ROC. The results, in Figs. 14–15, demonstrate a significant decrease in privacy leakage for DP models compared to non-DP models. The non-DP models exhibited an average AUC-ROC of 0.696, indicating an evident risk of privacy leakage. Conversely, the DP models showed a much lower AUC-ROC of 0.530, suggesting a significant reduction in MIA attack effectiveness. Finally, the differential privacy budget (ϵ) values indicate that while privacy is improved, the high values suggest that additional tuning is possible to achieve better privacy guarantees.

From a deployment perspective, both SA and DP introduce non-negligible overhead. SA increases communication rounds and requires additional cryptographic operations, especially in multi-level setups. DP, in turn, impacting training time. However, these costs are often acceptable in privacy-critical applications such as biometric systems in regulated environments.

4.3. Qualitative evaluation of risk assessment via DPIA

While the quantitative evaluation conducted through privacy auditing tools provides essential insights into privacy risks, it does not capture all aspects of data protection. To complement this analysis, we employ a DPIA, which offers a structured and standardized approach to qualitatively assessing privacy risks. This analysis is conducted in accordance with the methodology evaluated by the *Commission nationale de l'informatique et des libertés (CNIL)*⁷ (French national data protection and privacy agency), employing the open-source PIA tool that was developed by the CNIL itself.

The DPIA is a well-established framework defined by the GDPR to assess the potential risks associated with data processing activities. This assessment allows us to systematically evaluate the impact of different mitigation strategies, both current and future, on privacy.

The DPIA was conducted focusing on technical aspects, progressively integrating mitigation strategies aligned with the scenarios described in Section 3. While organizational factors were excluded, the DPIA remains a valuable tool to assess the effectiveness of privacy-preserving mechanisms and their impact on risk reduction.

⁷ <https://www.cnil.fr>

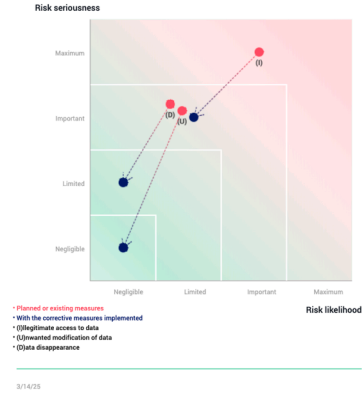


Fig. 16. Risk mapping differences.

The DPIA process requires a methodical evaluation of privacy issues through four main steps. First *Context Definition* specifies data categories of gathered data, involved participants and legal and regulatory requirements. Second, *Fundamental Principal Assessment* evaluates compliance with GDPR, focusing on minimization, proportionality and necessity of data processing. Third, *Risk Identification and Evaluation*, involves detecting potential privacy risks like access, modification, or loss, evaluated by severity and likelihood. Finally, *Risk Mitigation and Validation*, consists of implementing security controls and mitigation strategies, which are validated using risk mapping and comparative analysis.

The incremental approach allows us to evaluate privacy risks in three architectural scenarios, applying incremental risk mitigation measures.

4.3.1. Comparative risk analysis and risk mapping

The risks mitigation measures was applied progressively in the DPIA process highlighting how different levels of data protection measures impact privacy on the three different scenarios.

In the baseline scenario, the basic basic security controls were enforced, including encryption at rest, role-based access control, secure server training, and long-term data retention policies.

To reduce the risk associated with the centralized architecture, in the one-level FL architecture we adopted FL itself, secure aggregation, optional HE, and reduced data retention.

In the final step of applying privacy protection measures, we introduced DP. This is the main additional privacy-preserving characteristic of the two-level FL architecture, but not the unique one. In Table 3, all these measures are summarized with remarks of progression across the three reference architectures.

The final result of the privacy assessment is summarized in Table 4 in terms of severity and likelihood for each scenario and privacy risks. Fig. 16 shows the risk mapping assessment with the differences between the first scenario (red dots) and the final scenario (blue dots) with all the mitigation measures applied. All the privacy risks reduce their seriousness and likelihood and they are placed in the “green” area, which means “medium risk” according to of GDPR.

5. Results and discussion

Compared to the existing literature reviewed in the Section 2 that proposes FL frameworks combining HE and DP, they typically lack quantitative privacy audits and do not explicitly address decentralized training in multi-stakeholder environments. In contrast, our approach integrates both quantitative and qualitative evaluations, including privacy audits based on MIA using the Privacy Meter tool and a DPIA methodology. To the best of our knowledge, this is one of the first studies to systematically compare centralized, one-level, and two-level FL architectures for facial recognition in airport scenarios under privacy constraints, providing a complete view of the trade-offs between privacy and model performance.

Table 3
Mitigation measures across architectural scenarios.

Scenario	Mitigation Measure	Brief Description
Centralized	Data encryption at rest	Biometric and passenger data are encrypted at the database level to prevent leakage.
	Role-based access control	Data access is restricted through clearly defined roles and permissions.
	Secure model training on central servers	Models are trained in controlled environments, introducing a single point of failure.
	Data Retention Policies	Long-term data storage improves accuracy but increases the risk of data breaches.
One-level FL	Federated Learning	Clients train local models and share only aggregated updates, protecting raw data.
	Secure Aggregation for FL	A privacy-preserving protocol prevents access to individual client updates.
	HE	Model updates remain encrypted during aggregation, increasing computational overhead.
Two-level FL with DP	Shortened data retention for passenger info	Passenger data is stored only for essential operational purposes, reducing exposure.
	Differential privacy for model updates	Noise is added to model updates before aggregation to prevent data reconstruction.
	Gradient clipping	Limits the influence of single data points on the model to reduce leakage risk.
	Differential Privacy accounting	Tracks cumulative privacy loss ensuring the system does not exceed privacy budgets over time.
	Differential privacy in model deployment	Noise is included in the deployed model to prevent biometric pattern extraction.

Table 4
DPIA risk summary across three scenarios.

Risk Category	Metric	Scenario 1 (Centralized)	Scenario 2 (FL - No DP)	Scenario 3 (FL + DP)
Illegitimate Access to Personal Data	Severity	4	3	3
	Likelihood	3	3	2
Unwanted Modification	Severity	3	2	2
	Likelihood	2	1	1
Disappearance of Personal Data	Severity	3	2	1
	Likelihood	2	2	1

The privacy evaluation highlights a trade-off: while the centralized model achieves high accuracy, it remains highly vulnerable to MIA, as confirmed by elevated AUC-ROC values. This underscores the need for privacy-preserving techniques such as FL and DP.

In the second experiment, FL distributes training across multiple edge devices, preventing the centralization of data. This decentralization reduced data exposure, but the privacy audit results showed that model updates were still sensible for membership inference. Specifically, the results from the two-client configuration indicate that MIA remains effective. However, increasing the number of nodes reduced the MIA effectiveness, as suggested by the five-nodes experiment. Thus, increasing the number of FL nodes improves privacy-preserving but it also highlights a limitation of the only FL approach: it does not completely prevent inference attacks. Therefore, additional privacy-preserving mechanisms could be needed. For these reasons, in the third architecture, more sensitive to privacy loss, DP was introduced to further mitigate the risk of inference attacks. DP injects noise into the model before model updating, reducing the success rate of MIA. These results are shown by the lower value of AUC-ROC resulting in the analysis. The effectiveness of DP in improving privacy preservation is widely demonstrated, but at a high-performance cost. Indeed, the noise added during training resulted in a decrease in recognition accuracy, highlighting the fundamental trade-off between privacy and accuracy. Although this evident drawback, in biometric systems, privacy protection is paramount, and the reduction in accuracy may be a necessary compromise.

The qualitative analysis conducted through the DPIA confirmed the results of the quantitative experiments. The three architectures show decreasing exposure to privacy risks as mitigation measures are applied. The first architecture shows the highest level of exposure to all the risks identified by the analysis, the second one mitigates the risk of large-scale data loss thanks to the distribution of data across nodes; and finally, the third architecture shows the highest level of privacy protection. The risk mapping also confirms this progression towards the highest level of privacy protection, as the results in Table 4 and Fig. 16 show. The DPIA

has demonstrated its value in assessing these improvements, providing a methodology for assessing privacy risks beyond purely quantitative metrics. Although this study focused primarily on technical risks, a real-world DPIA would also need to address organizational and legal aspects such as compliance monitoring and governance policies. Nevertheless, the DPIA integration within Privacy Meter auditing provides concrete evidence of the benefits and trade-offs associated with FL and DP in biometric systems.

6. Conclusion and future works

The study highlights FL and DP potential to enhance privacy in facial recognition systems and the need for careful consideration of trade-offs and comprehensive risk management strategies. Some key conclusions drawn are the following. Centralized facial recognition systems, while operationally efficient and accurate, pose significant privacy risks, as demonstrated by their vulnerability to MIA. High AUC-ROC values in analysis reveal that the system is prone to data exposure. FL is a promising solution by decentralizing data storage and processing, thereby reducing privacy risks. FL alone is insufficient, as model updates can still be exploited for inference attacks, particularly in smaller networks with fewer nodes. The two-level FL architecture further improves model generalization and privacy preservation introducing vulnerabilities, such as model poisoning risks, necessitating additional mitigation strategies like DP. DP effectively counters inference attacks by injecting noise into model updates, as evidenced by lower AUC-ROC values in the analysis. However, reducing accuracy, highlighting the inherent trade-off between privacy and performance. Increasing the number of nodes in a FL network enhances privacy resilience by diluting the impact of individual contributions, but this scalability also introduces computational overhead, which must be carefully balanced against privacy gains. The DPIA framework complements quantitative analysis by providing a structured methodology to assess privacy risks, confirming that privacy protection improves incrementally with the adoption of FL and DP. The DPIA also emphasizes the need to address organizational and legal aspects, such as compliance monitoring and accountability mechanisms, in real-world implementations. The study demonstrates that a hybrid approach, with FL and DP, offers the highest level of privacy protection, albeit with some performance trade-offs. This approach is particularly critical in sensitive applications like biometric authentication, where privacy is paramount. While the study focuses on technical risks, a comprehensive DPIA in real-world scenarios must also consider organizational, legal, and governance factors. Future research should aim to optimize the balance between privacy and utility, especially in large-scale environments like airports, and explore advanced privacy-preserving techniques to address emerging threats like model poisoning will be crucial for the safe deployment of biometric systems.

Moreover, future evaluations will include system-level metrics such as latency and bandwidth to better assess the feasibility of real-world deployments.

CRedit authorship contribution statement

Helio Campanile: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Conceptualization; **Maria Stella de Biase:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Conceptualization; **Fiammetta Marulli:** Writing – review & editing, Writing – original draft, Conceptualization.

Data availability

Public dataset.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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